
Data: Current data D ; historical data $D_0 = \{D_{0h} : h = 1, \dots, H\}$;
number of posterior samples M ; hyperparameters ξ .

Result: M posterior samples under the RMAP prior.

- 1 Obtain M posterior samples under the BHM using the current and historical data, $p_I(\boldsymbol{\theta}|D, D_0)$.
 - 2 Obtain M posterior samples under a vague prior using the current data only, $p_V(\boldsymbol{\theta}|D)$.
 - 3 Compute the normalizing constants $Z_I(D, D_0)$ and $Z_V(D)$ via bridgesampling, then compute $\tilde{\gamma}$ in (14).
 - 4 For each $m = 1, \dots, M$, draw an indicator $j_m \sim \text{Bernoulli}(\tilde{\gamma})$. If $j_m = 1$, sample (with replacement) $\boldsymbol{\theta}^{(m)}$ from the posterior samples corresponding to $p_I(\cdot|D, D_0)$; otherwise, sample (with replacement) $\boldsymbol{\theta}^{(m)}$ from those corresponding to $p_V(\cdot|D)$.
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